

SIH 2026 · PS ID 26192

PRALAYADARSHI

Hyperlocal Flash Flood & Landslide Early Warning System

Predict Early. Alert Locally. Save Lives.

Problem Statement ID: 26192

• Ministry of Home Affairs

• NDRF — Disaster Management Division

THE PROBLEM

Why Current Warning Systems Fall Short



Warnings cover large regions, not individual villages.



Remote areas have poor communication infrastructure.



Flash floods develop within minutes or hours.



Authorities lack real-time evacuation intelligence.



Important data sources work separately.

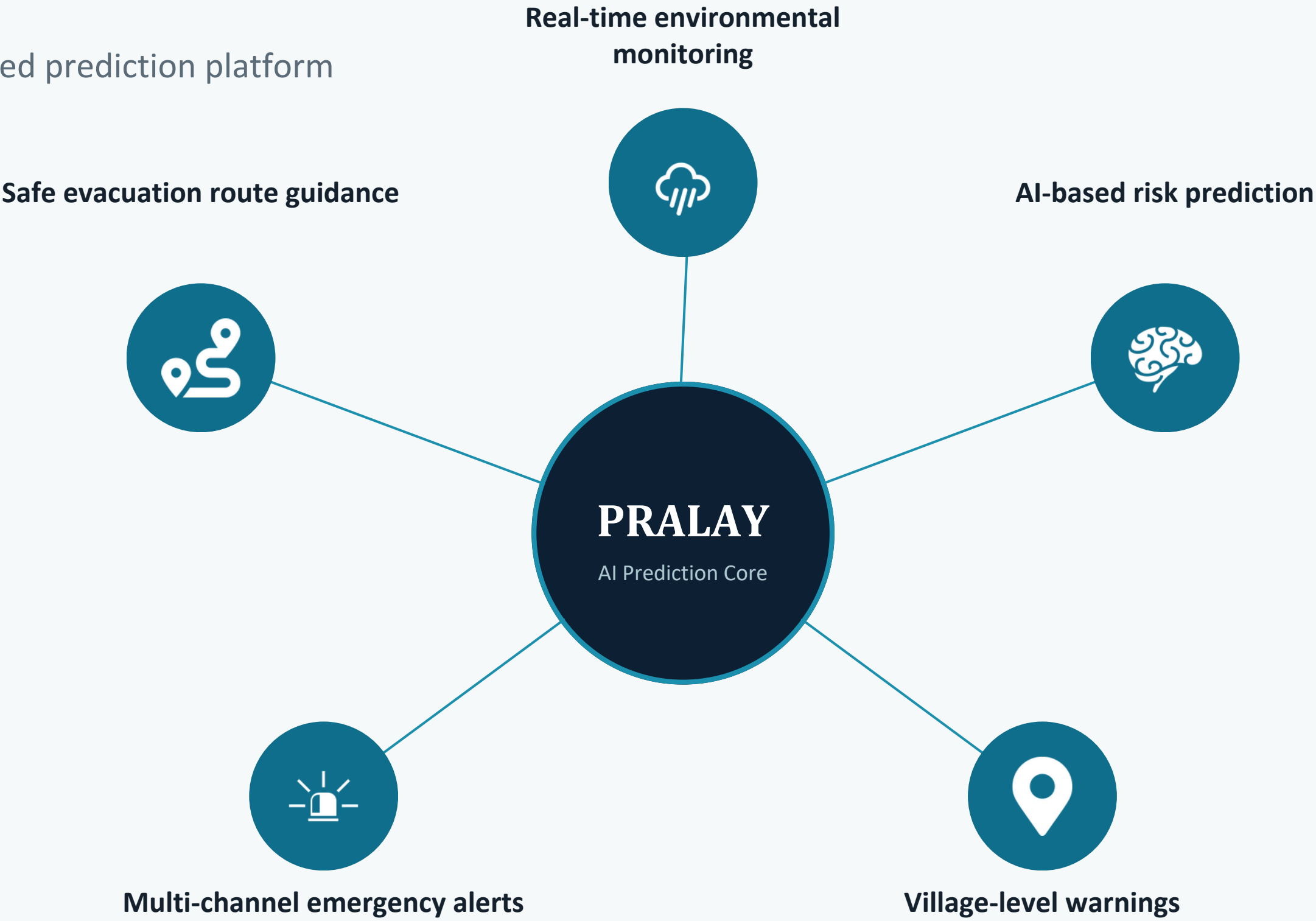


A warning is useful only if it reaches the right person before disaster strikes.

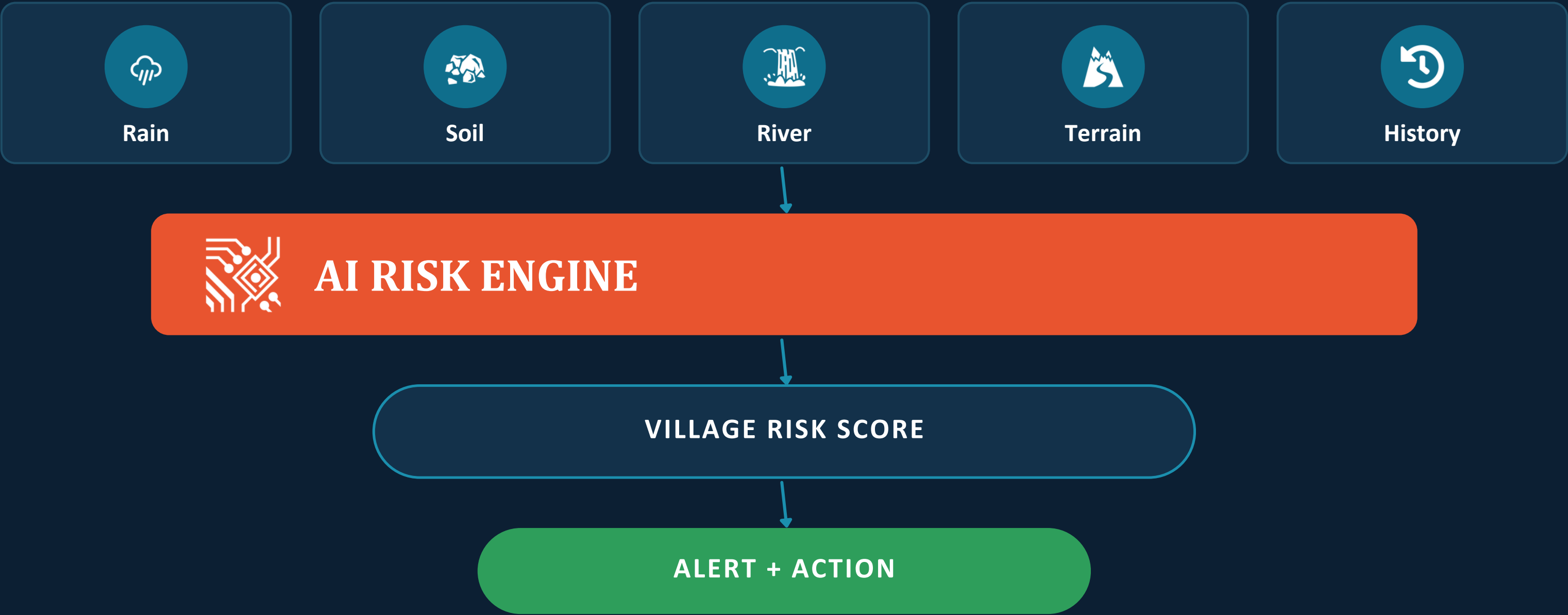
OUR SOLUTION

Introducing PRALAYADARSHI

A multi-source AI-powered prediction platform



Multi-Source Intelligence Pipeline



Weather data · IoT sensors · Satellite imagery · GIS terrain · Historical disaster records

From Raw Data to Early Warning



Flood Risk

Based on:

- Rainfall intensity
- River water rise
- Drainage capacity
- Terrain elevation



Landslide Risk

Based on:

- Soil moisture
- Slope angle
- Rainfall duration
- Historical incidents

OUTPUT

Low

Moderate

High

Critical

Hyperlocal Risk Map



TAP A VILLAGE TO REVEAL

-  Disaster probability
-  Current conditions
-  Lead time
-  Recommended action

Smart Evacuation & Alert System

CITIZENS



Mobile alerts



SMS



Sirens



Voice calls

AUTHORITIES



Risk dashboard



Resource allocation



Shelter availability



Dynamic safe routes

What Makes Us Different?

Beyond Just Prediction

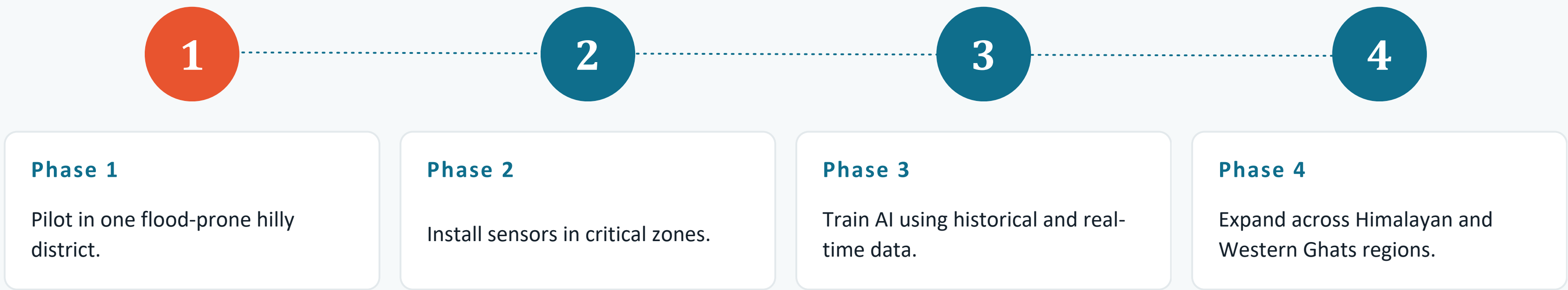
TRADITIONAL SYSTEM

-  District-level alerts
-  Single data source
-  Static warnings
-  Warning only
-  Manual monitoring

PRALAY

-  Village-level alerts
-  Multi-source intelligence
-  Real-time risk updates
-  Warning + evacuation
-  AI-assisted monitoring

Implementation & Scalability



POTENTIAL PILOT LOCATIONS

-  Himachal Pradesh
-  Uttarakhand
-  Sikkim
-  Arunachal Pradesh
-  Maharashtra Western Ghats

Every Minute of Warning Can Save Lives



Provide earlier warnings



Improve evacuation planning



Support NDRF and local authorities



Reduce casualties



Protect infrastructure



Build disaster-resilient communities

"We cannot stop natural disasters. But we can make sure people are not caught unprepared."